



FORMAT FOR COURSE CURRICULUM

Course Title: Introduction to Python

Credit Units: 2

Course Level: UG

Course Code: ECE319

L	T	P/S	SW/FW	No. of PSDA	TOTAL CREDIT UNITS
1	-	2	-		2

Course Objectives:

- Learn Syntax and Semantics and create Functions in Python
- Handle Strings and Files in Python
- Understand Lists, Dictionaries and Set in Python
- Implement Object Oriented Programming concepts in Python

Pre-requisites: Student should be familiar with fundamental programming constructs in any programming language like C, C++ etc.

Course Contents/Syllabus:

	Weightage (%)
Module I	15
Introduction: Features, Installation of Python, Basic Syntax, Variable and Data Types, Operators	
Module II	20
Conditional Statements: If, If- else, Nested if-else, Looping: For, While, Nested loops, Control Statements: Break, Continue, String Manipulation: Accessing Strings, Basic Operations, File Operation, File Reading, File Writing, Appending File	
Module III	25
Functions: Defining a function, Calling a function, Types of functions, Function Arguments, Global and local variables Functions	
Module IV	25
Class and object, Attributes, Inheritance Lists: Introduction, Accessing list, Operations, Working with lists, Function and Methods, Tuple: Introduction, Accessing tuples, Operations, Functions and Methods	

Module V	15
Dictionaries: Introduction, Accessing values in dictionaries, Working with dictionaries, Properties, Functions, Set : Introduction, Accessing Values in a Set, Operations on Set	

Course Learning Outcomes:

- Student will be able to examine Python syntax and semantics and be fluent in the use of Python flow control and functions
- Student will be able to demonstrate proficiency in handling Strings and File Systems
- Student will be able to create, run and manipulate Python Programs using core data structures like Lists, Dictionaries and use Regular Expressions.
- Student will be able to interpret the concepts of Object-Oriented Programming as used in Python
- Student will be able to implement exemplary applications related to Network Programming, Web Services and Databases in Python

Pedagogy for Course Delivery:

1. Chalk and Talk
2. LCD/SMART Boards
3. Web Resources
4. Student Seminars
5. Guest Lectures

List of Professional Skill Development Activities (PSDA): NA

Lab/ Practicals details, if applicable:

- Write a program to read two integers and find the sum, diff, mult and div.
- Write a program to find the simple interest for a given value P, T and R. The program must take the input from the user.
- Write a python program to generate first n Fibonacci number and factorial of n using functions
- Write a python program to count the number of Characters in a given string and hence to display the characters of string in reverse order. • Write a python program to
To create two new files f1 and f2
To read and display the contents , count the number of lines and find the word whose count is more in f1 and f2 respectively.
To create and display the file f3 which is a combination of f1 and f2.
- Write a Python program to convert a list of characters into a string.
- Write a Python program to find the index of an item in a specified list.
- Write a Python program to get the frequency of the elements in a list.

- Write a Python program to convert a list of tuples into a dictionary
- Write an Object-Oriented Python program to create two Time objects: *current Time*, which contains the current time; and *bread Time*, which contains the amount of time it takes for a bread maker to make bread. Then we'll use add Time to figure out when the bread will be done. Write the print Time function to display the time when the bread will be done by the bread maker.

Assessment/ Examination Scheme:

Theory L/T (%)	Lab/Practical/Studio (%)
50	50

Theory Assessment (L&T):

Continuous Assessment/Internal Assessment (40 %)					End Term Examination (60%)
Components (Drop down)	Mid-Term Exam(CT)	Class Quiz	Home Assignment	Attendance(A)	
Linkage of PSDA with Internal Assessment Component, if any					
Weightage (%)	15	10	10	5	70

Lab/ Practical/ Studio Assessment:

	Continuous Assessment/Internal Assessment (40 %)				End Term Examination (60 %)	
Components (Drop down)	PR	LR	V	A	Practical	Viva
Weightage (%)	15	10	10	5	30	30

A: Attendance, PR- Performance, LR – Lab Record, V – Viva. EE- External Exam

Text Reading:

- Charles R. Severance, “Python for Everybody: Exploring Data Using Python 3”, 1st Edition, Create Space Independent Publishing Platform, 2016.
- Allen B. Downey, "Think Python: How to Think Like a Computer Scientist”, 2ndEdition, Green Tea Press, 2015.
- Reema Thareja, “Python Programming Using Problem Solving Approach” Oxford university press, 2017

References:

- Charles Dierbach, "Introduction to Computer Science Using Python", 1st Edition, Wiley India Pvt Ltd. ISBN-13: 978-8126556014
- Mark Lutz, “Programming Python”, 4th Edition, O’Reilly Media, 2011.ISBN-13: 978-9350232873
- Wesley J Chun, “Core Python Applications Programming”, 3rd Edition, Pearson Education India, 2015. ISBN-13: 978-9332555365
- Roberto Tamassia, Michael H Goldwasser, Michael T Goodrich, “Data Structures and Algorithms in Python”, 1st Edition, Wiley India Pvt Ltd, 2016. ISBN-13: 978- 8126562176

Additional Reading:

- NPTEL
- SWAYAM

Any other Study Material:

- TUTORIALS POINT